

PROVINCIAL DATA BANK SYSTEM

Database Architecture and

Dimensional Model Design

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1. Introduction

This document presents the **proposed database design for the Provincial Data Bank System** of the Provincial Government of Bataan. The system's goal is to organize and store data related to various provincial indicators reported by different agencies, offices, and departments.

The objective of the **Proposed Data Bank System** is to create a centralized data storage without needing to perform different requests or communications with various offices and waiting for approval before getting the data source to perform data analysis or monitoring for success indicators related to the **Government Joint Strategic Goals**.

Tracking the indicators is crucial to measure the success of **different PPAs (Program, Projects, Activities)**. Using the **Data Bank System** as a data source for not only for the success indicator but other strategic goals of departments to prevent unnecessary complexities regarding getting the right data.

The structure is represented through an **Entity Relationship Diagram (ERD)**, which demonstrates the entities as tables, their attributes, and relationships required to support the analysis of provincial indicators.

1.1 Purpose of the Data Bank System

The Proposed Data Bank system aims to create a centralized repository of data related to indicators across provincial agencies. By unifying data sources into a structured repository, the system enables easier data analysis, reporting, auditing, and data-driven decision making within the Provincial Government of Bataan.

1.2 Scope of the Design

This report focuses on the **conceptual** and **physical database design** of the Proposed Data Bank System, particularly the schema structure and dimensional modeling approach implemented to organize the data for analyzing the provincial indicators.

Implementation aspects such as the development of data pipelines, system deployment, application interfaces, etc. are outside the scope of this document and may addressed in future phases.

2. Database Design Approach

The proposed design implements the **dimensional modeling approach**, more commonly used in analytical systems and data warehouses to answer different questions by aggregating and summarizing data.

Dimensional modeling organizes the data into **fact tables** and **dimension tables**, allowing easier storage and analysis of indicators across multiple perspectives of reporting.

This modeling approach was selected because the **Proposed Data Bank** primarily supports monitoring and reporting where data must be analyzed across different dimensions such as time, date, location, implementing offices, etc.

2.1 Fact Tables

Fact tables capture **quantitative** data or metrics of a certain business process such as counts, rates, and percentages. By capturing this type of data, it can support the foundation of data analysis to be used for reporting success indicators.

Examples of fact tables in the proposed schema include:

- fact_prices
- fact_crime_reports
- fact_accident_reports
- fact_participation_rates

2.2 Dimension Tables

Dimension tables capture **qualitative** data that stores descriptive attributes that provides context to the fact table. The values of this table are typically geographic locations, dates, time periods, organizational units, or even government sectors.

Examples of dimension tables in the proposed schema include:

- dim_dates
- dim_times
- dim_public_enterprises

These dimensions allow indicator data to be analyzed across different perspectives that provides references to the fact table to support stronger data analysis.

3. High-Level Entity Relationship Diagram

To demonstrate the clear overview of the structure of the database, a **high-level ERD** is presented below.

The current schema contains more than **70 tables and exactly 37 indicators**, the overview diagram focuses on the **core structural components** of the system.

The diagram includes:

- Fact tables representing provincial indicators
- Global shared dimensions used across multiple indicators
- Partially shared dimensions used by several indicators

Indicator-specific dimensions are excluded from this overview in order to preserve readability.



Figure 1. High-Level Entity Relationship Diagram of the Proposed Provincial Data Bank

Legend:

- **Royal Blue** – Fact Tables
- **Golden Orange** – Global Shared Dimensions
- **Teal** – Multi-Indicator Dimensions
- **Gray** – Limited-Shared Dimension

4. Example Indicator Modules

Because of the size of the diagram proposed as a design for the **Data Bank System**, several modules are presented to demonstrate how different indicators are structured within the dimensional model.

Each module typically consists of:

- one or two **fact table** containing measurable values of one indicator
- several **dimension tables** providing contextual information of the **fact table**

4.1 Inflation Monitoring Indicator

Indicator: Reduction percentage in inflation rates through price monitoring and direct selling of basic commodities.

This module captures the data for the price observations of agricultural commodities across different municipalities of **Bataan**.

The fact table stores the average and prevailing prices, while the dimension tables provide contextual information

Tables involved:

- fact_prices
- dim_local_government_units
- dim_agricultural_products
- dim_price_comparison_periods
- dim_dates

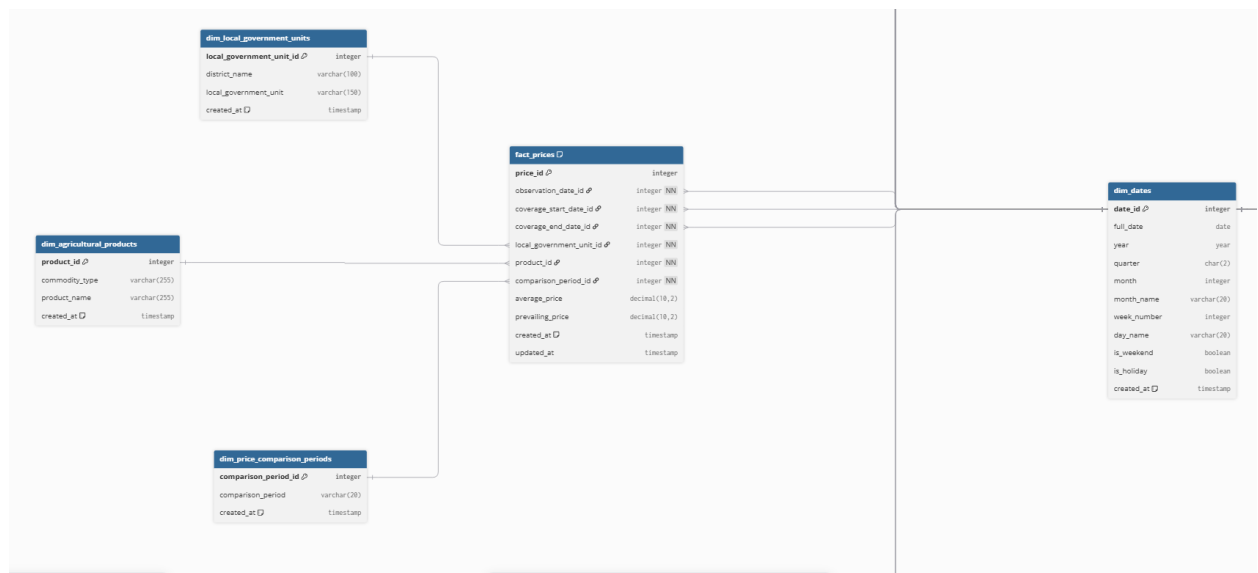


Figure 2. Dimensional Model for the Inflation Monitoring Indicator

4.2 Crime Monitoring Indicator

Indicator: Reduction percentage in crime rates through collaboration with law enforcement agencies.

This module captures the crime data reported to the police stations located across different municipalities of **Bataan**.

The fact table records the number of index crime reported and other metrics, while dimension tables identify the police station when the index crime is reported and its date and time. Index crimes are prioritized to monitor because they represent serious offenses and affect the public safety.

Tables involved:

- fact_crime_reports
- dim_police_stations
- dim_dates
- dim_times

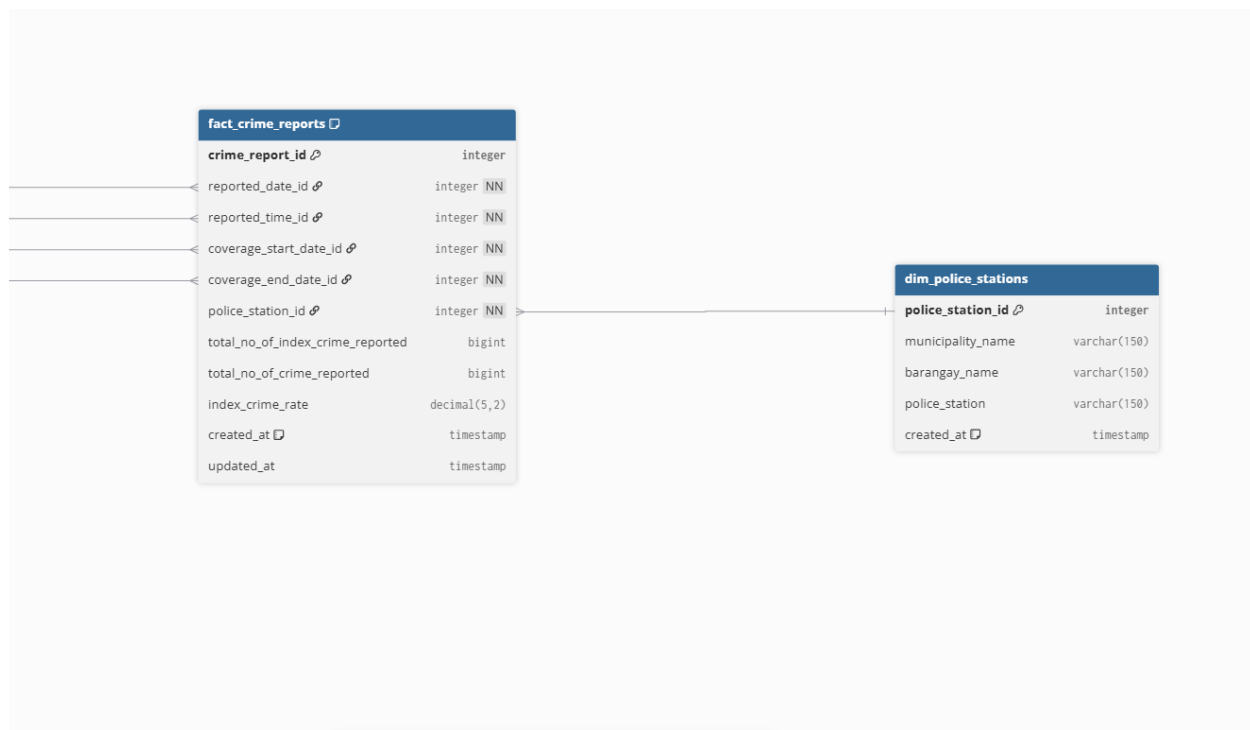


Figure 3. Dimensional Model for the Crime Monitoring Indicator

5. Table Categories

The tables inside the database can be classified into several categories based on their role within the schema.

Category	Description	Example Tables
Global Shared Dimensions	Dimensions used across all indicators	dim_dates
Multi-Indicator-Specific Dimensions	Dimensions used by more than 2 indicators but not all	dim_times
Limited-Shared Dimension	Dimensions used only by 1-2 indicators	dim_public_enterprises
Fact Tables	Tables storing quantifiable data	fact_prices

6. Guide to Exploring the Full ERD

To match the high-level ERD in **Section 3**, the **complete database schema** is available through multiple formats for a more detailed exploration.

Available Access:

- **Full Entity Relationship Diagram Image:** [Click here](#)
- **Interactive Entity Relationship Diagram:** [Click here](#)
- **Github Repository:** [Click here](#)

Quick Guide to the Interactive Entity Relationship Diagram

1. Open the dbdiagram.io link
2. Zoom and pan to explore the schema
3. Hover over a column of a certain table to see its relationships
4. Hover over a fact table to identify which indicator it belongs to

Note: Edit privilege is not available due to the current subscription plan, preventing also accidental changes to the layout of the interactive **Entity Relationship Diagram**.

7. Future Implementation Considerations

While this report focuses on the **database schema design**, different components may be required to fully implement the Proposed Data Bank System.

Future development may include:

- Development of **data ingestion pipelines** to collect data from different provincial departments
- Implementation of **ETL/ELT processes** for cleaning and transforming data
- Establishment of **data validation** and **quality** monitoring mechanisms
- Implementation of **data governance policies** across offices
- Development of a website to create a **unified interface** for government employees to access data from different sources.

These future implementation considerations would allow the Data Bank System to evolve into a **fully operational data platform.**

8. Conclusion

This proposed Data Bank System establishes a scalable and structured foundation for centralized provincial data management, to enable efficient monitoring of multiple government indicators and supporting data-driven decision making across departments.

By implementing a dimensional modeling approach, the system enables faster and efficient storage, retrieval, and analysis of a given indicator across various reporting dimensions such as **date, time, location, and responsible offices.**

The combination of the entity relationship diagram, interactive schema, and DBML source code allows reviewers to understand and assess the feasibility of implementing the **Data Bank System for the Provincial Government of Bataan.**